

5G/6G Communication Systems- ECA5601 (LAB)

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Experiment 7: Analog Beamforming Analysis Using MATLAB

Aim:

To analyze the performance of analog beamforming by studying beamforming gain, beam width,

and main lobe direction using MATLAB.

Objectives:

1. To analyze the beamforming gain achieved by varying the number of antenna elements in an analog beamforming system using MATLAB.
2. To compare the beam width of an antenna array for different beamforming configurations using MATLAB.
3. To analyze the main lobe direction by steering the antenna array beam to different angles using MATLAB.

Software Required:

- MATLAB

Objective 1:

Program:

```
clc;
clear;
close all;

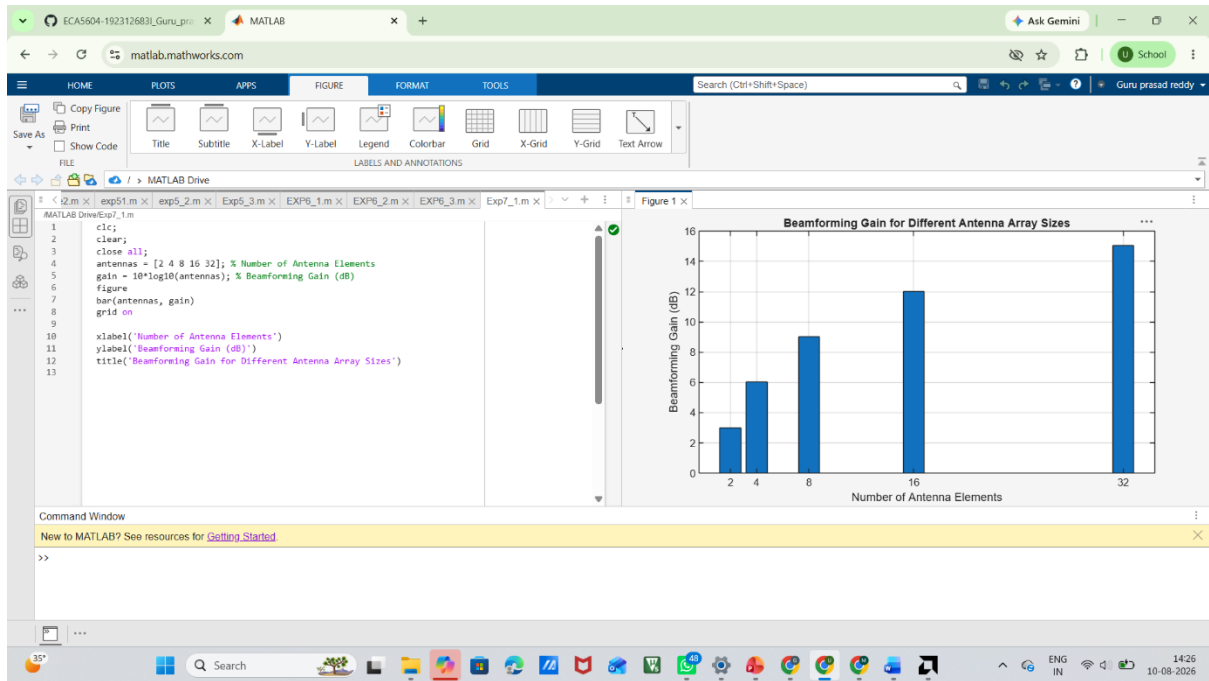
antennas = [2 4 8 16 32]; % Number of Antenna Elements
gain = 10*log10(antennas); % Beamforming Gain (dB)
figure
bar(antennas, gain)
grid on
```

```
xlabel('Number of Antenna Elements')
```

```
ylabel('Beamforming Gain (dB)')
```

```
title('Beamforming Gain for Different Antenna Array Sizes')
```

Output:



Objective 2:

Program:

```
clc;
```

```
clear;
```

```
close all;
```

```
antennas = [2 4 8 16 32]; % Number of Antenna Elements
```

```
beamwidth = [90 45 22.5 11.25 5.625]; % Beam Width (degrees)
```

```
figure
```

```
plot(antennas, beamwidth,'o-','LineWidth',2)
```

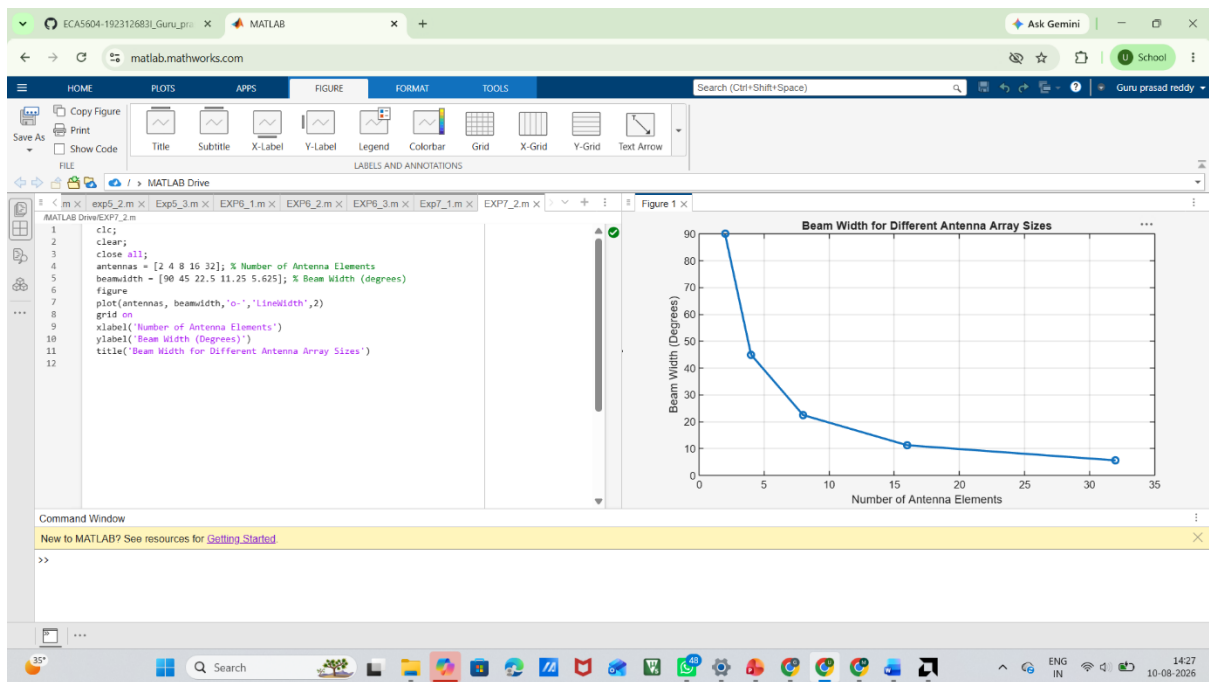
```
grid on
```

```
xlabel('Number of Antenna Elements')
```

```
ylabel('Beam Width (Degrees)')
```

```
title('Beam Width for Different Antenna Array Sizes')
```

Output:



Objective 3:

MATLAB Code:

```
clc;
```

```
clear;
```

```
close all;
```

```
theta = [-60 -30 0 30 60]; % Main Lobe Direction (degrees)
```

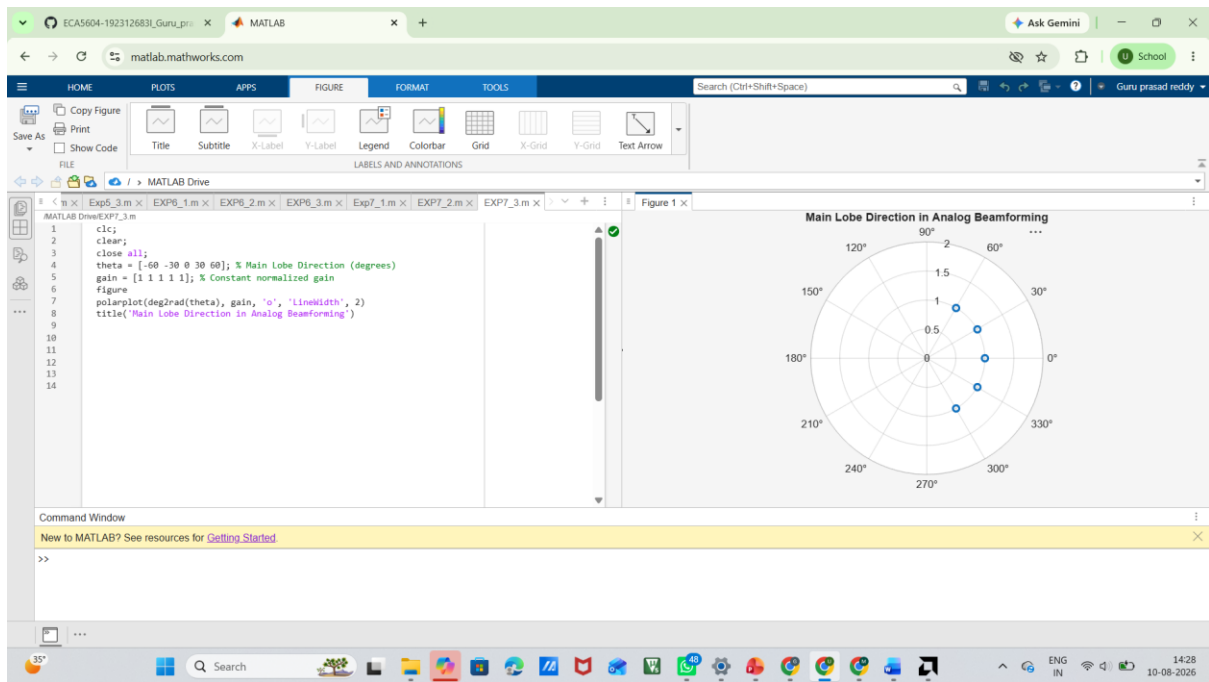
```
gain = [1 1 1 1 1]; % Constant normalized gain
```

```
figure
```

```
polarplot(deg2rad(theta), gain, 'o', 'LineWidth', 2)
```

```
title('Main Lobe Direction in Analog Beamforming')
```

Output:



Result:

The MATLAB simulation showed that increasing the number of antenna elements improves the beamforming gain and produces a narrower beam width. It also confirmed that the main lobe can be accurately steered to different directions, demonstrating the effectiveness of analogy beamforming.